

Deliberation and Learning: Evidence From Deliberative Polls

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Abstract

Among the great virtues ascribed to deliberation is learning. We ask how much participants in Deliberative Polls learn, whether taking part causes the learning, and who learns. In four Deliberative Polls with control groups, held in seven countries, participation raised factual knowledge by .46 standard deviations (95% interval .34 to .58); in the one poll followed for a year, 45% of the effect remained. Across 22 polls without controls, the share of items answered correctly rose by .130 [.099, .162], about twice as much face to face as online. Participants who start out knowing less, and those without degrees, gain somewhat more from participation. Small groups are composed as if at random, and who is in one's group, including how much the others know, has no detectable effect on learning.

1 Introduction

Among the great virtues commonly ascribed to deliberation is learning. The participants learn about the topics under discussion and, to a lesser degree, incidentally, about related ones. They learn not only relevant facts but the arguments for and against particular alternatives and what other people unlike themselves, their circumstances, and their ways of thinking are like. This is helpful, both at the individual level, for discerning what policies most closely accord with one's values and interests and voting accordingly, and at the aggregate level, for forming "authentic" majorities—majorities that "know what they are doing" (Luskin 2003). How much citizens know shapes what they prefer and how they vote (Converse 1964; Bartels 1996; Althaus 1998; Delli Carpini and Keeter 1996), even on foreign policy (Aldrich et al. 1989; Foyle 2004), though information shortcuts can partly stand in for knowledge (Popkin 1991; Lupia 1994).

The expectation of learning follows directly from the nature of "deliberation," conventionally construed as discussion, but not just any discussion (Fishkin and Luskin 2005). While definitions vary, deliberation is generally viewed as normatively elevated discussion—in which, among other things, the participants clarify and refine their preferences by open-mindedly weighing competing arguments. As part of that weighing, they absorb factual information and sift out misinformation.

Or so the theory runs. But how much do participants actually learn, is it the deliberation that teaches them, and who learns the most? We examine learning of relevant facts. This is not the only sort of learning involved, but it is the easiest to measure, and it is both important in itself and positively correlated with the rest.

Deliberative Polls teach. In four polls that surveyed a control group alongside the participants—two in the United States, one in Tanzania, and one across six middle-income countries—taking part raised factual knowledge by .46 standard deviations of the controls' initial knowledge, and in the one poll re-interviewed a year later, 45% of that effect remained. Earlier studies of Deliberative

Polls and other forums found that participants learn a lot (Luskin et al. 2002; Barabas 2004; Sturgis et al. 2005; Esterling et al. 2011; Theuwis et al. 2025), but most compared participants before and after, without a comparison group. We add two things. Participation helps those who start out knowing less somewhat more. And because small groups are composed as if at random, group composition is a natural experiment, and it shows no effect of one’s groupmates on how much one learns.

2 Deliberative Polling and Learning

Deliberative Polling (Fishkin 1991, 1995; Luskin et al. 2002) aims to capture fuller-information, more thoughtful opinions than those expressed in ordinary polls. In very brief, a Deliberative Poll interviews and assembles, as many as are willing of, a random sample to deliberate some policy or electoral choice, in alternating small group discussions and plenary question-and-answer sessions with panels of competing experts, and then re-measures the sample’s opinions at the end. These latter are frequently different, both at the individual level and in the aggregate, from what they were before deliberating. The absolute net change is statistically significant nearly three-quarters of the time and averages nearly .1 on a 0–1 scale (Luskin et al. 2007).

This design, admittedly a particular embodiment of deliberation, affords additional opportunities and incentives to learn. The participants are provided with carefully balanced briefing documents, rehearsing the arguments each side would make. There is usually a section of basic background information, and the arguments embed each side’s pet facts—those most congenial to its arguments. In addition, the participants know from the moment they are first recruited that they will be attending the deliberation and that the deliberation (typically) will be an “Event,” capital “e”—away from home, usually televised or taped for later broadcast, often attended by high-profile policy experts or policy-makers in the role of observers or panelists. Thus, motivated by a desire to “do well,” many participants use the weeks between their recruitment and the event itself to prepare—not only by reading the briefing materials but by paying heightened attention to stories about the topic in the media, by talking more with friends, family, and coworkers about it, and even, in some cases, by researching it in the library or online.

So if knowledge, like its close relative “sophistication,” depends on the confluence of cognitive ability, motivation, and exposure to relevant information (Luskin 1990, 2002), the deliberation in Deliberative Polling stands to increase two of those three. There is not much to be done, in adulthood, about ability—and in fact we should want the participants to be representative in this, as in all other respects. But the treatment can and does affect motivation and exposure. Judging from past polls, that is enough to produce considerable, sometimes dramatic learning (Luskin et al. 2002; List et al. 2013).

The online design is intended to be as similar as possible to the face-to-face one, given the difference in mode. The deliberation is by voice rather than text, so as to capture as much as possible of deliberation’s social elements. The participants still “meet” in small groups, still led by trained moderators, and still question panels of policy experts and policy makers. They still receive briefing materials. They are still a representative sample—sometimes randomly sampled offline, with “nonusers” receiving free access, including free equipment, to get them online, sometimes drawn from online panels and matched to approximate a random sample. On the other hand, deliberating by voice, even by video, online can never capture the full richness and complexity of in-person interactions; the discussions are interrupted, spread over a period of weeks, with much greater possibility of real-world intrusion, and the assignment to small groups is perforce constrained by

possible meeting times. All these differences may act to weaken the online treatment’s effects (Luskin et al. 2006; Iyengar et al. 2005).

3 Data

We use two kinds of data (Table 1; Appendix A lists every poll). For 22 polls held between 1994 and 2009 we have each participant’s proportion of knowledge items answered correctly before (T1) and after (T2) deliberating, with small-group membership and demographics: 21 from the replication data for Luskin et al. (2022), and a poll in Marousi, Greece, in which a party chose its mayoral candidate, released here for the first time. For 23 polls, most of them the same, we have responses item by item (Cor and Sood 2016). Four more recent polls surveyed control groups.

Table 1: Data. Participants and controls are panel respondents used in the analyses. Control groups are an uninvited random sample, or randomized; in the antimicrobial-resistance poll the treated group is the attendees.

Study	Design	Participants	Controls
21 Deliberative Polls, 1994–2009	Pre-post	5,867	
Marousi, Greece, 2006	Pre-post	146	
23 Deliberative Polls, item level	Pre-post	6,654	
America in One Room, 2019	Uninvited sample	523	844
America in One Room: Climate, 2021	Uninvited sample	962	671
Tanzania, 2015	Randomized	383	921
Antimicrobial Resistance, 2024	Randomized; attendees	1,280	1,139

The control groups differ in how they were formed. America in One Room and its 2021 climate successor interviewed a separate random sample that was never invited (Fishkin et al. 2021). The Tanzanian poll randomized villages to treatment or control and, within treated villages, individuals to deliberation or to the briefing materials alone (Sandefur et al. 2022). The poll on antimicrobial resistance, held online in Brazil, Colombia, India, Indonesia, Nigeria and Tanzania, randomized panel members after the first interview; the release contains the controls and the invitees who attended, but not the invitees who stayed home (Mendelson et al. 2026). Appendix Table 4 lists our recodes of the pre-post data.

4 Measuring Learning

We measure knowledge by closed-ended knowledge questions, mostly multiple choice, in some cases extracted from party or candidate placements, and in a few cases true-false (Luskin 1987; Luskin and Bullock 2005; Price 1999; Mondak 1999, 2001). Appendix D lists the items. Correct responses are scored as 1, incorrect responses and “don’t knows” as 0 (Luskin and Bullock 2011), and these scores are then averaged across items to give the proportion answered correctly.

We must also confront a distinction between knowledge gain ($K_2 - K_1$) and post-deliberation knowledge (K_2), where K is observed (measured, as distinct from actual) knowledge, subscripted for wave of measurement. Let us denote the respondent’s actual knowledge at these same times as K_1^* and K_2^* , his or her actual knowledge gain (presumed to be nonnegative) as $\Delta K^* = K_2^* - K_1^*$, and his or her observed knowledge gain as $\Delta K = K_2 - K_1$. K_2 is a better proxy for ΔK^* than

ΔK is. Indeed, under hard-to-assail assumptions, ΔK can be *negatively* correlated with ΔK^* and is generally less highly correlated with ΔK^* than K_2 is (Luskin et al. 2016). The intuition is that the participants who know a lot at the outset (a) cannot show much increase in knowledge, because they already have the maximum score or something close to it, but (b) are those who are actually learning the most, consistent with the rich-get-richer thrust of the relevant literatures in psychology and communication research (Kinder 1998). ΔK will be very low for precisely the participants for whom ΔK^* is highest. K_2 , by contrast, is high for all those learning the most, either because they started low and observably learned a lot or because they started (and remained) high and unobservably learned a lot. So when explaining who learns we model K_2 with K_1 among the regressors—what Allison (1990) called the “regressor variable” as opposed to the “change score” approach—and when describing how much a poll’s participants learned we report the mean gain.

Closed-ended items invite guessing, which inflates K_1 and can hide learning. The latent class model of Cor and Sood (2016) separates, item by item, respondents who knew the answer, learned it, or guessed, assuming nobody forgets. Across the 23 polls with item-level data, its estimates of learning average .208, against a raw .159.

5 How Much Do Participants Learn, and Does Participating Cause It?

Figure 1 puts all the evidence on one scale, standard deviations of T1 knowledge. Panel A shows the mean gain in each of the 22 polls without a control group; panel B shows the effect of participation in the four polls with one.

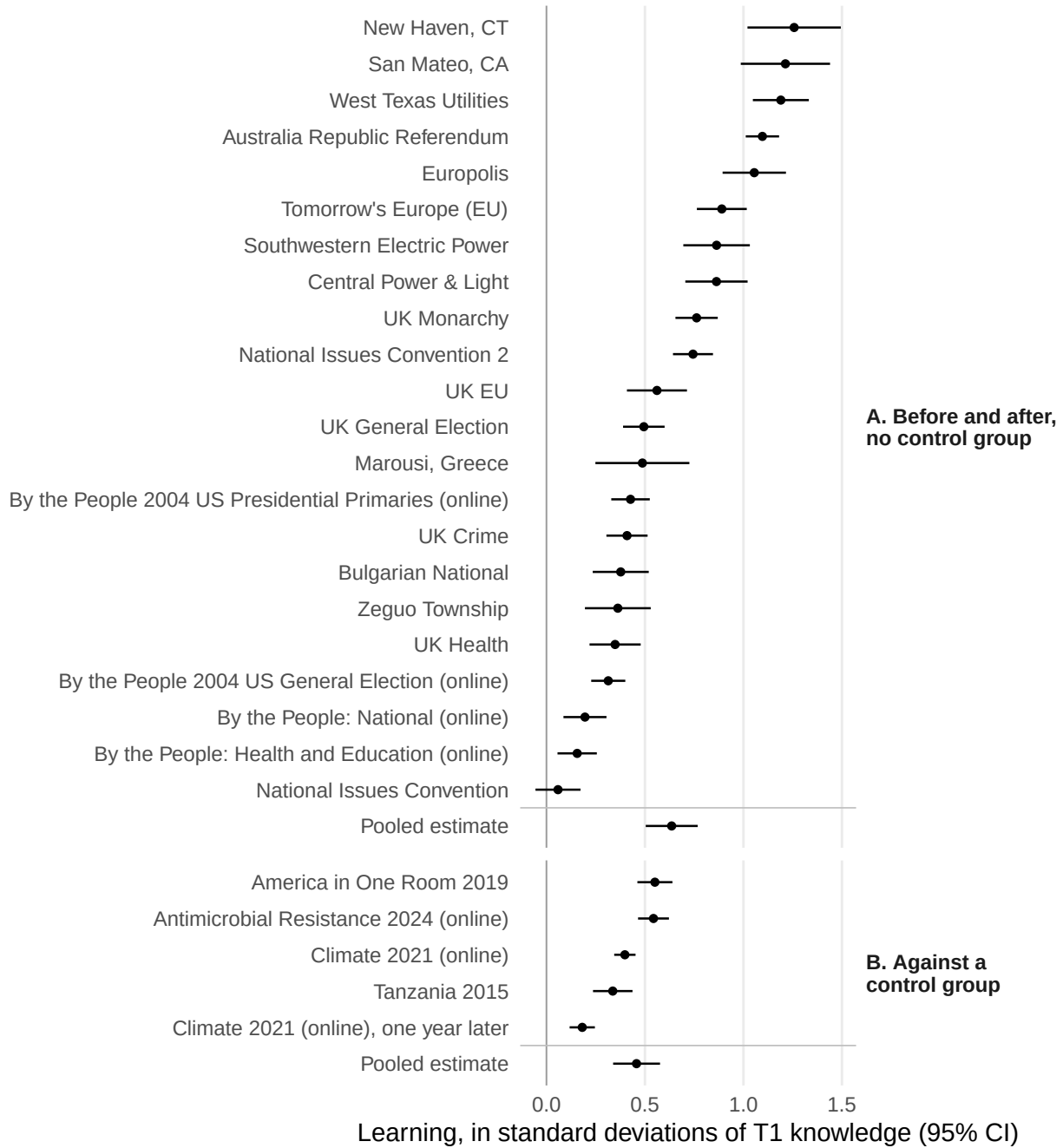


Figure 1: Learning in every poll, in standard deviations of T1 knowledge, with 95% intervals. A: mean T2 minus T1 gain among participants, scaled by the poll's T1 standard deviation. B: effect of participation, from a regression of T2 on treatment and T1 knowledge, scaled by the control group's T1 standard deviation; Tanzanian standard errors are clustered by village. Pooled estimates are from Bayesian random-effects meta-analyses.

Every poll shows learning. Raw gains range from .012 to .288 of the items; pooled, the proportion correct rose by .130 [.099, .162], or .64 [.50, .77] standard deviations. The pooling is a Bayesian random-effects meta-analysis: each poll's mean gain is its true gain plus sampling error, true gains are normally distributed around an overall mean, the prior on the mean is flat and that on the

between-poll standard deviation half-normal with scale .1 (Röver 2020). Face-to-face polls produced about twice the gain of online ones (.145 [.113, .177] against .065 [-.001, .132]; Appendix Table 5).

A pre-post gain alone does not show that deliberation taught anything: participants take the same quiz twice and live through weeks of news. The control groups net these out, and participation still teaches. Adjusting for T1 knowledge, it added .142 (.012) to the proportion correct in America in One Room, .121 (.008) in the 2021 climate poll, and .125 (.009) in the antimicrobial-resistance poll, where the country estimates range from -.04 in Tanzania to .31 in Nigeria. In Tanzania’s national poll, deliberation raised a standardized knowledge index by .335 (.051) standard deviations, matching the published estimate (Sandefur et al. 2022). Pooled, the effect is .46 [.34, .58] standard deviations. A year after the climate poll, attendees still knew .055 more than controls (s.e. .010), 45% of the immediate effect. The controls’ own gains, .01, .03 and .02 of the items in the three studies scored that way, show how little of a pre-post gain testing and news explain.

The designs differ in what they rule out. Tanzania randomized; its estimates are causal. The antimicrobial-resistance poll randomized too, but its treated group is the attendees, who were 23 to 92% of invitees depending on the country, so the comparison adds self-selection to randomization. The two American polls compare attendees with a separately drawn sample, and attendees are a selected subset of invitees: in America in One Room they answered .458 of T1 items correctly against .378 for invitees who stayed home, and in the climate poll .686 against .579. Conditioning on T1 knowledge removes differences in where people start. The difference in gains, which instead assumes that the two groups would have gained equally, gives .131, .093 and .103 in the three studies; when both assumptions are plausible the truth lies between the two estimates (Angrist and Pischke 2009).

Does deliberation add anything to the briefing materials? In Tanzania, deliberation and the briefing materials alone were randomized among individuals in the same villages. Deliberation added .205 (.058) standard deviations adjusting for T1 knowledge, but only .081 (.081) in the difference of gains, because the deliberation arm happened to start .16 standard deviations higher. Discussion probably helps; this one comparison cannot say by how much.

6 Who Learns?

6.1 Regressors and hypotheses

Our explanatory variables vary by level of aggregation—some individual-level, some group-level, some poll-level—and substantively, in the degree to which they revolve around motivation versus ability versus exposure to relevant information. All are measured before deliberation.

Initial knowledge (K_1). The information-rich tend to get information-richer. The more one knows going in, the more one tends to learn and thus know going out. Absent ceiling effects, the coefficient should be > 1 ; given them, it may be < 1 . Prior knowledge on a particular topic improves people’s ability to comprehend new information, enabling them to extract the central elements of a message and draw appropriate inferences efficiently (Eckhardt et al. 1991; Recht and Leslie 1988). Prior knowledge also enhances people’s ability to store new information on that topic and retrieve it later (Cooke et al. 1993; Fiske et al. 1990; Hambrick 2003; McGraw and Pinney 1990; Recht and Leslie 1988; Schneider et al. 1993). So the less political information individuals have stored in memory, the more difficult it is for them to acquire new information.

Education. Education, qua years of schooling, is distinct from, and only moderately correlated with,

political knowledge. Controlling for cognitive ability, with which it is more strongly correlated, its effect on knowledge appears negligible (Luskin 1990). But absent a separate measure of cognitive ability, it can roughly proxy the latter. Naturally, we expect it to have a positive effect. There may, however, be some mutual substitution with initial knowledge. People who know enough already may not need to be as bright to learn a lot, and people who are bright enough may not need to know as much to do so. We measure education as comparably as possible across polls in three categories: less than high school graduation, high school graduation, and a bachelor’s degree or more.

Gender and age. These demographic variables exhibit real-world political knowledge gaps, with men knowing more than women and older people more than younger ones (Delli Carpini and Keeter 1996). Age is measured in decades and gender as a dummy scored 1 for women.

Proportion of women. Whether from greater comfort, more widely shared cognitive or communication style, or lower stereotype threat (Steele 1997), women may learn more in small groups containing more women. We calculate the proportion excluding the participant herself.

Mode. Online deliberation, with its thinner exposure to other participants, may be, either intrinsically or at least in its current instantiations, a somewhat etiolated version of the face-to-face treatment (Luskin et al. 2006). It is therefore plausible that people learn less from online deliberations.

Group size. The number of people in the participant’s small group may affect the quality of the deliberation, with larger groups arguably making for a less informative experience. As group size increases, attention may wander, and participants may not learn all that they hear (Latané et al. 1979).

Group knowledge. Participants may learn more from the other members of their small group when the latter know more. This suggests using the mean knowledge of the group’s other members. But what should matter is not how much the other group members know but how much they know about the particular facts the participant does not already know. So where the item-level data allow, we also measure group knowledge as the other members’ mean T1 score on the items the participant answered incorrectly at T1.

Heterogeneity of opinion. In the real world, people often know congenial but not uncongenial facts—those commonly cited by those sharing their views but not those commonly cited by those holding opposing views. Part of deliberation’s effect on learning may consist in getting people to note and accept (if, in many cases, still argue around) uncongenial facts. This can be expected to occur mainly on site. Participants learn a great deal in anticipation of the event, but the learning at that stage is mainly of congenial facts (Hansen 2004). On site, the participants find it harder to avoid confronting the uncongenial ones. This may happen more, the more opposing points of view are represented and argued in the small group—the more attitudinally heterogeneous the group is. We gauge this as the generalized variance, the determinant of the covariance matrix, of the group’s policy attitudes (Anderson 2003).

Attitude extremity. Participants starting with more extreme views may have shuttered out more uncongenial facts, leaving them with more ground for deliberation to make up. We measure extremity as the mean, across attitude indices j , of $|A_{ij} - .5|$, where A_{ij} is the i th respondent’s score on the j th index.

Reading the briefing materials. One of the main information vectors is the briefing document sent to all participants shortly after recruitment, normally well in advance of the on-site deliberations.

It provides some background facts and lays out the major arguments for and against given policy options, which embed additional facts congenial to each argument. For four polls we have the participants' reports of how much of the briefing document they read, which we score linearly and use in a separate analysis confined to those polls.

Issue salience. More salient issues, to which many participants have already given some attention, exhibit decidedly less net attitude change (Luskin et al. 2007) and fall shorter of single-peakedness (List et al. 2013). We suspect they also, and relatedly, show less learning: there is less left to learn. We proxy salience by the participants' mean T1 knowledge.

6.2 Model

The nested structure of the data, with individuals clustered in small groups clustered in polls, suggests a hierarchical linear model (Raudenbush and Bryk 2002) with random intercepts by small group and poll:

$$K_{igd2} = \beta_0 + \alpha_g + \gamma_d + \mathbf{x}'_{igd}\beta + u_{igd},$$

where i , g and d index individuals, small groups and polls; $\alpha_g \sim N(0, \sigma_\alpha^2)$ and $\gamma_d \sim N(0, \sigma_\gamma^2)$ are unmodeled parts of the intercept; $\mathbf{x}_{igd}$ holds the regressors above; and $u_{igd} \sim N(0, \sigma_u^2)$ is a disturbance. We estimate it by maximum likelihood (Bates et al. 2015). The main model uses 5,827 participants in 412 groups in 22 polls.

6.3 Whom participation helps most

Before turning to the model, the control-group studies let us ask whose knowledge participation raises most, a causal question the pre-post data cannot answer. Adding the product of treatment and centred T1 knowledge, the effect of participation is smaller for those who start knowing more, by -.15 (.04) per unit of T1 knowledge in America in One Room and -.25 (.03) in the climate poll, with no difference in the other two studies (Appendix Table 6). Part of this is the ceiling: those who start high have less room to show a gain. Participation also helped those without a bachelor's degree somewhat more in the climate and antimicrobial-resistance polls (-.047 (.016) and -.043 (.020)), though not in America in One Room. Deliberation narrows knowledge gaps modestly; it does not widen them.

6.4 Who ends up knowing the most

Table 2: Post-deliberation knowledge: maximum likelihood estimates (standard errors) with random intercepts for small group and poll. The second column is limited to the 8 polls whose item-level answers link to small groups, which allows group knowledge to be measured on the items each participant missed at T1. The third is limited to the 4 polls that asked about the briefing materials and uses poll fixed effects in place of poll-level variables.

	All polls	Item-specific (8 polls)	Briefing (4 polls)
Intercept	.053 (.049)	.070 (.069)	.519 (.095)
T1 knowledge	.482 (.025)	.583 (.047)	.515 (.124)
High school	.122 (.016)	.210 (.027)	.078 (.048)
BA or more	.195 (.018)	.275 (.029)	.158 (.048)
T1 knowledge x high school	-.052 (.029)	-.218 (.052)	-.088 (.131)
T1 knowledge x BA or more	-.102 (.032)	-.266 (.055)	-.125 (.129)
Age (decades)	-.007 (.002)	-.009 (.003)	-.014 (.004)
Attitude extremity	-.007 (.018)	.019 (.020)	-.121 (.057)
Group size	.000 (.001)	.001 (.003)	-.005 (.003)
Group knowledge: others' mean T1	.086 (.051)		-.205 (.114)
Group knowledge: others' T1 on items missed		-.058 (.044)	
Opinion heterogeneity	.024 (.082)	.002 (.123)	.118 (.164)
Female	-.025 (.023)	-.093 (.040)	-.049 (.047)
Proportion female in group	-.011 (.034)	-.071 (.056)	-.047 (.068)
Female x proportion female	.030 (.044)	.140 (.072)	.069 (.088)
Online	-.099 (.033)	-.148 (.047)	
Poll mean T1 knowledge	.517 (.104)	.568 (.119)	
Read briefing materials			.065 (.016)
Participants	5,827	2,108	1,289
Small groups	412	150	98
Polls	22	8	4

Table 2 shows the estimates. T2 knowledge depends heavily on T1 knowledge (.482 (.025)); measurement error in T1 scores pulls this estimate toward zero. Education also has a powerful effect: a bachelor's degree adds .195 (.018) for a participant who starts with nothing right. Education and prior knowledge substitute for one another, as the negative interaction indicates (-.102 (.032) per unit of T1 knowledge): among the well educated, those who initially know the least close much of the gap. Older participants end knowing slightly less (-.007 (.002) per decade). Participants in online polls end up knowing less (-.099 (.033)), and those in polls whose participants start out knowing more end up knowing more (.517 (.104)), contrary to the salience hypothesis. In the four polls that asked, those who read more of the briefing materials knew more afterwards (.065 (.016)), though reading is itself a choice. Attitude extremity and opinion heterogeneity have no detectable effect.

6.5 Do small groups matter?

If participants learn from each other, those placed with more knowledgeable groupmates should learn more. Estimating this requires that groupmates' characteristics be unrelated to one's own, which random assignment to groups ensures and self-selection does not. Organizers intend random

assignment. We test it by regressing each participant's characteristic on the mean among the other members of the group, controlling for the mean among the other members of the poll, without which the test is biased toward finding negative sorting (Guryan et al. 2009). The coefficients are near zero for T1 knowledge (.017 (.017)), gender (-.016 (.020)) and age (-.008 (.025)), and small for a bachelor's degree (.042 (.030)). Groups are composed as if at random.

That makes group composition a natural experiment. Within each of 23 polls, the 22 above and America in One Room, we regress T2 knowledge on the other members' mean T1 knowledge, controlling for one's own and the poll's, and pool the estimates (Figure 2). The pooled effect is .069 [-.047, .190]: at the upper end, a group whose other members answered ten more points of the T1 items correctly would raise one's T2 score by 1.9 points. The share of women in the group has no detectable effect either (.027 [-.013, .066]). Measured on the items each participant missed, which is closer to what one could learn from others, group knowledge again has no effect (-.058 (.044) in the 8 polls with linked item data). Group knowledge measured after the deliberation would show an effect for the wrong reason: groupmates' T2 answers reflect the experience they shared with the participant (Manski 1993).

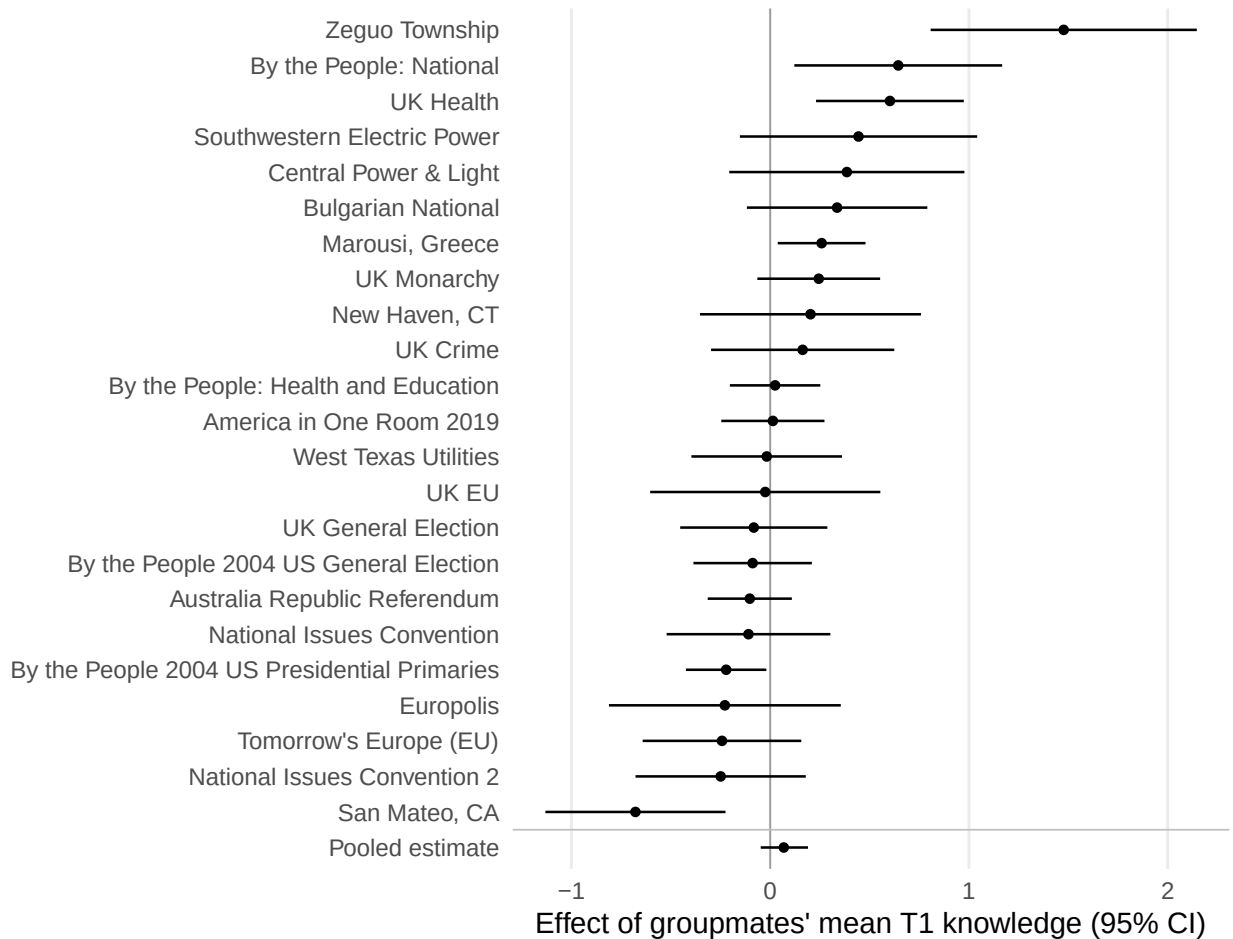


Figure 2: Effect of the other group members' mean T1 knowledge on own T2 knowledge, by poll, with 95% intervals. Each estimate controls for own T1 knowledge and the poll's leave-one-out mean, with standard errors clustered by small group; the pooled estimate is from a Bayesian random-effects meta-analysis.

7 Conclusion

Deliberative Polls raise what participants know by about half a standard deviation, and much of the gain lasts. The learning does not depend on whom participants happen to deliberate with, and it favours, if anything, those who start out knowing less. The simplest reading is that the whole event teaches, through the briefing materials, the expert sessions, the discussion, and the anticipation of taking part, rather than through particular groupmates.

Two limits remain. Only Tanzania randomized people to take part or not; elsewhere the comparison involves attendees, who differ from those who stay home. And the knowledge items are few and chosen by the organizers, so the gains are gains in the facts the event emphasized.

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A The Deliberative Polls

Table 3: The Deliberative Polls without control groups. Data: P, participant-level scores; I, item-level responses.

Poll	Year	Topic	Data	Source
UK Crime	1994	Crime	P+I	Luskin et al. (2002)
UK EU	1995	Britain’s future in Europe	P+I	Sturgis et al. (2005)
UK Monarchy	1996	Future of the monarchy	P+I	Sturgis et al. (2005)
National Issues Convention	1996	Foreign and economic policy	P+I	Fishkin and Luskin (1999)

Poll	Year	Topic	Data	Source
Central Power & Light	1996	Electricity choices	P+I	Luskin, Fishkin, and Plane (1999)
West Texas Utilities	1996	Electricity choices	P+I	Luskin, Fishkin, and Plane (1999)
Southwestern Electric Power	1996	Electricity choices	P+I	Luskin, Fishkin, and Plane (1999)
UK General Election	1997	Electoral choice	P+I	Luskin, Fishkin, Jowell, et al. (1999)
UK Health	1998	National Health Service	P+I	Park et al. (1998)
Australia Republic Referendum	1999	Becoming a republic	P+I	Luskin et al. (2000)
Denmark: Euro	2000	Adopting the euro	I	Hansen (2004)
National Issues Convention 2	2003	Foreign policy	P	Luskin et al. (2006); <i>By the People</i> (2003)
By the People: National (online)	2003	Foreign policy	P	Luskin et al. (2006)
By the People 2004 US Presidential Primaries (online)	2004	Presidential primaries	P	Iyengar et al. (2005)
By the People 2004 Online Primaries (online)	2004	Presidential primaries	I	Iyengar et al. (2005)
By the People 2004 US General Election (online)	2004	Presidential election	P+I	
New Haven, CT	2004	Airport and regional tax sharing	P	Farrar et al. (2010)
By the People: Health and Education (online)	2005	Health care and education	P+I	<i>By the People</i> (2005)
Zeguo Township	2005	Local infrastructure projects	P	Fishkin et al. (2010)
Marousi, Greece	2006	Choosing a mayoral candidate	P	Center for Deliberative Democracy (2006)
Tomorrow's Europe (EU)	2007	Pensions, globalization and enlargement	P+I	Luskin et al. (2008)
Bulgarian National	2007	Roma housing, crime and education	P+I	Center for Deliberative Democracy (2007)
Northern Ireland	2007	Local education	I	Fishkin et al. (2007)

Poll	Year	Topic	Data Source
Vermont Energy	2007	Energy choices	I
By the People 2007 (online)	2007	Electoral reform	I
San Mateo, CA	2008	Housing	P+I Threshold 2008 and Center for Deliberative Democracy (2008)
Europolis	2009	Climate change and immigration	P+I
Michigan	2009	Economy and budget	I
California What's Next	2011	Government reform	I

B Recodes

Table 4: Recodes applied to the participant-level data

Poll	Variable	Rule	Rows affected	Evidence
National Issues Convention	mode	online 1 -> 0	all rows	Held face to face in Austin, January 1996
National Issues Convention	ppage	age = 1996 - ppage	all rows	ppage ranges 1900-1989 in this poll only, i.e. birth year
All polls	ppage	age < 16 or > 100 -> NA	Zeguo age 1; Europolis age 109	Deliberative Polls sample adults
BTP 2004 Presidential Primaries	all	exact duplicate rows dropped	217 of 434 rows	Exact duplicates in the public release
Marousi, Greece	t2know	0 -> NA when T1 > 0	26 of 146 rows	27 T2 scores of 0 against 2 at T1, implausible for a knowledge scale after deliberation
Marousi, Greece	educ3	educ4 code 7 -> NA	20 rows	Code outside the four education categories
All polls	t1know, t2know	rounded to 10 decimals	2 NIC rows	Stored as 1.0000000002

C Further results

Table 5: Bayesian random-effects meta-analysis of poll mean gains, in proportion correct

Model	Parameter	Median	2.5%	97.5%
Pooled	Mean gain	.130	.099	.162
Pooled	Between-poll SD	.070	.050	.096
By mode	Face-to-face	.145	.113	.177
By mode	Online	.065	-.001	.132
By mode	Between-poll SD	.064	.045	.088

Table 6: Heterogeneous effects of participation. Each row adds the product of treatment and the moderator to the regression of T2 on treatment and T1 knowledge.

Study	Moderator	Treatment x moderator	n
America in One Room 2019	T1 knowledge (centred)	-.155 (.041)	1,367
America in One Room 2019	Bachelor’s degree	.027 (.023)	1,367
America in One Room: Climate 2021	T1 knowledge (centred)	-.252 (.029)	1,633
America in One Room: Climate 2021	Bachelor’s degree	-.047 (.016)	1,633
Antimicrobial Resistance 2024	T1 knowledge (centred)	.043 (.038)	2,419
Antimicrobial Resistance 2024	Bachelor’s degree	-.043 (.020)	2,408
Tanzania 2015	T1 knowledge (centred)	-.018 (.051)	1,304

D Knowledge items

The items for the polls in the participant-level data, with correct answers in parentheses where the item is not multiple choice.

D.1 UK (EU)

(All questions are true/false)

- EU has recently expanded to 15 members
- Switzerland is to join EU (false)
- Britain’s income tax rates are decided in Brussels (false)
- Elections to European Parliament are held every 5 years
- Of the three British parties, Liberal Democrats least in favour of EU.

D.2 UK (Health)

(All questions are true/false)

- Standard charge for NHS prescription is 10 pounds (false)
- Proportion of old people in Britain getting larger (true)
- Even with inflation taken into account, spending on NHS doubled over last 20 years (true)
- Most people use private medical treatment instead of NHS (false)
- All British women get free breast cancer screening on NHS (false)

- Cosmetic surgery is never available on NHS (false)

D.3 UK (GE)

(First three questions are true/false. Others are party placement on a 7 point scale.)

- Inflation has been less than 5% over the past few years (true).
- Interest rates are decided by Bank of England (false).
- Unemployment in UK is higher than in Germany (false).

(Conservative, Labor, Liberal Democrat) Party's View: Government should do nothing to make incomes in Britain more equal.

(Conservative, Labor, Liberal Democrat) Party's View: Government should spend much less on services like education and health in order to cut taxes a lot.

(Conservative, Labor, Liberal Democrat) Party's View: Government should definitely not introduce a minimum wage because too many low paid workers would then lose their jobs.

(Conservative, Labor, Liberal Democrat) Party's View: Britain should do much more to keep its distance from the European Union.

D.4 AU (Monarchy)

Issue Related Items

- Which one of these best describes the current role of the Queen in relation to the Australian Governor General? - She chooses the Governor General, She appoints the Governor General only on the advice of the Prime Minister, She appoints the Governor General only on the recommendation of parliament, She has no role in choosing or appointing the Governor General
- Which one of these is true of the current role of the Governor-General in Australia – He performs only ceremonial duties, He can decide whether or not to dismiss the government, He acts only on the Queen's instructions, He controls the government (He performs only ceremonial duties)
- If the referendum on the republic is passed, which one of these would the role of the President be more like? – The current Governor General, The American President, The British Prime Minister, the current Australian Prime Minister (The current Governor General)
- If the referendum on the republic is passed, which one of these best describes how the Australian Prime Minister in the Republic can remove the President? - At any time without reporting to parliament, At any time but must later obtain approval from the House of Representatives, Only after a fair trial in parliament, The Prime Minister could not remove president.

Under the proposed change to a Republic, which of these, as far as you know, would definitely change.

- The Australian Flag (will not change)
- The National Anthem (will not change)
- The word 'Royal' in the names of the Royal Australian Navy and the Royal Australian Air Force (will change)
- Australia's participation in the Commonwealth Games (will not change)

- On the whole would you say the Liberal Party, or the Labor Party, is more in favour of Australia becoming a Republic?

General Politics Items

- Which one of these best describes Aden Ridgeway - Leader of the federal opposition, Leader of the Australian Democrats, An aboriginal senator in parliament, A justice of the high court, Or is he none of these
- Which one of these best describes Jennie George - A Labor member of federal parliament, Secretary of the Teacher's Federation, President of the ACTU, Leader of the Australian Worker's Union, Or is she none of these
- And now about social and welfare issues. On the whole would you say the Liberal Party, or the Labor Party, is more concerned about social and welfare issues, they are equally concerned, or do you not have much of an impression about that?
- And now thinking about business. On the whole would you say the Liberal Party, or the Labor Party, is closer to business, they are equally close or do you not have much of an impression about that?

D.5 UK (Crime)

(All questions are true/false)

Issue Related Items

- British courts are allowed to sentence a murderer to death (false).
- Britain has largest prison population in Western Europe (true).
- Britain has more people serving life sentences than rest of European Community put together (false).
- Possible to be tried by jury in local magistrate court (false).

General Politics Items

- Number of MPs about 100 (false).
- Longest time allowed between general elections is 4 years (false).
- Britain's electoral system is based on proportional representation (false).

D.6 UK (Monarchy)

(All questions are true/false)

- Princess Anne next in line to throne. (false)
- PM, not the queen, is Britain's Head of State. (false)
- Queen is head of Commonwealth. (true)
- Queen's duty to decide election date. (false)
- No PM takes office without being asked by the Queen. (true)
- Queen is head of Church of England. (true)
- Queen is head of Church of Scotland. (false)
- Britain has unwritten constitution (true).

D.7 EU 2007

- Which of the following countries is an official candidate to join the EU? - Romania, Montenegro, Croatia, Morocco (Croatia)
- Are the members of the European Parliament – Directly elected by the citizens of their country, Elected for three year terms, Elected by the parliament of their country, Appointed by their national head of government (directly elected by the citizens of their country)
- Are new EU laws in the field of employment adopted by - The European Commission, in some cases with the European Parliament, The Council
- of Ministers, in some cases with the European Parliament, The European Parliament by itself, The European commission by itself (The Council of Ministers, in some cases with the European Parliament)
- Is the EU's role regarding unemployment benefits to - Finance the member states' unemployment benefit systems, Decide the level and length of unemployment benefits in the member states, Require that the member states merge their unemployment benefit systems by 2010, Guarantee that all EU citizens have access to unemployment benefits where they live (Guarantee that all EU citizens have access to unemployment benefits where they live)
- Roughly a third of the EU's budget is devoted to one of the following. Is it ... ? - Helping the EU's less prosperous regions, Subsidizing the EU's fishing industry, Financing diplomatic missions abroad, Maintaining the EU's administration and bureaucracy (Helping the EU's less prosperous regions).
- By 2050, is the percentage of the adult EU population that is 65 or older projected to be - About one quarter of what it is now, About half of what it is now, About the same as what it is now, About twice what it is now, About four times what it is now (About twice what it is now)
- Which of the following is true of the amount of foreign aid given by the EU and its member states, combined, versus the amount given by the US? - The EU and its member states give roughly four times as much, The EU and its member states give roughly twice as much, The EU and its member states and the US give about the same amount, The US gives roughly twice as much, The US gives roughly four times as much (The EU and its member states give twice as much)
- On average, what percentage of the total spending by the governments of the EU member states for foreign aid? Is it - About 1%, About 5%, About 9%, About 13%, About 17% (About 1%)
- Which of the following countries does NOT possess nuclear weapons? - Pakistan, India, North Korea, Japan (Japan)

On a scale from 0 to 10, where 0 means as far left as possible, 10 means as far right as possible, and 5 is exactly in the middle, where would you place

- Nicholas Sarkozy (Greater than 5)
- Gordon Brown (Less than 5)

D.8 New Haven

- Which of the following would you say is closest to the population of the greater New Haven region? (550,000);
- During the 1990s, would you say the rate of job growth in the New Haven region was more, about the same, or less than the rest of the United States? (more);

- Which of the following would you say is the major source of revenue for most town governments in the Greater New Haven region? (Property taxes);
- Would you say the population of the city of New Haven increased, stayed the same, or decreased during the 1990s? (decreased);
- Does Connecticut law allow communities to share property tax revenues? (yes);
- Would you say those communities with the most valuable property tend to have the lowest property tax rates, average property tax rates, or the highest property tax rates? (lowest rates);
- Which of the following categories would you say the Federal Aviation Authority would use to describe Tweed New Haven Airport? (non-hub);
- Would you say that maintaining Tweed New Haven Airport at its current level of service would require any significant investment?(yes)

D.9 NIC 2

- Which of the following is closest to what you believe is true about global warming? (Caused by human activities like driving cars and burning fuel);
- Some people think that global warming is not really a problem so there is no need to do anything about it. Others think it's a serious problem and we need to act now.
- Where would you place the Democratic Party? ("It's a serious problems and we need to act now");
- Where would you place the Republican Party? ("Global warming is not really a problem so there is no need to do anything about it");
- As far as you know, does President Bush support or oppose recent international agreements to control greenhouse gases, or haven't you heard anything about this? (Oppose);
- As far as you know, does President Bush want to increase foreign aid, decrease foreign aid, or keep it the same, OR haven't you heard anything about this? (Increase);
- Out of every \$100 in federal budget, about how many dollars would you say goes to military spending? (\$20);
- Out of every \$100 in the federal budget, about how many dollars would you say goes to foreign aid? (\$1);
- For every \$100 in goods & services produced by the US, how many dollars worth would you say are sold to customers abroad? (\$12);
- In those African countries with the highest rates of infection with the AIDS virus, roughly how many adults out of every 100 would you say have AIDS or the AIDS virus? (30 or more);
- True or false - "The US has veto on WTO decisions" (False);
- True or false - "The US has veto on the UNSC" (True)

D.10 BTP03

(same as NIC2 questions)

D.11 BTP04

- How did John Kerry vote in the Senate on the resolution authorizing President Bush to go to war with Iraq? (Voted for);
- Which of the following countries now harbors the most Al Qaeda and Taliban fighters? (Pakistan);

- True or false - "Iraq was directly involved in the attacks on the WTC and the Pentagon on 9-11 2001" (False);
- True or false - "Large quantities of weapons of mass destruction have been found in Iraq" (False);
- True or false - "On average, prescription drugs cost more in Canada than in the US" (False);
- Which of the following was true of George W. Bush during the Vietnam War? (served in the TX Air National Guard);
- Which of the following was true of John Kerry during the Vietnam War? (decorated officer serving in Vietnam);
- A major destination of white collar jobs that have gone to other countries is? (India);
- Which of the following is closest to the number of Americans killed in Iraq since the war began? (1000)

D.12 San Mateo

- According to the Association of Bay Area Governments, between 1999 and 2006, what was the gap each year between the number of new homes needed and the number actually produced in San Mateo County? (About 1000);
- As of Sept 2007, about how much was the median price of a single-family home in San Mateo County? (\$950,000);
- As of 2006, about what percentage of County households could afford a median-priced house in San Mateo County? (12%);
- As of 2007 what was the average rent for a 2-bedroom apartment in San Mateo County? (\$1700);
- According to the 2006 San Mateo County Housing Needs Study, by about how many housing units is the demand for housing in San Mateo County projected to grow between 2005 and 2025? (70,000);
- According to the same Housing Needs Study, among the new households that are expected to form in the County between 2005 and 2025, about what percentage, combined, would be households with low, very low and extremely low-incomes? (About half);
- According to the 2006 San Mateo County Housing Needs Study, roughly how many new homes (including houses, townhomes, condos, and apartments) will be built between 2005 and 2025 at the current rate of housing development? (25,000);
- What percentage of San Mateo County's land is agricultural use, watershed, open space, wetlands, or parks? (more than 75%)

D.13 BTP05

- Among 29 wealthy industrialized nations, does the US rank top 3, top 10, bottom 10, or bottom 3 in students' math skills? (Bottom 10);
- In the last few years, it has become harder, easier or about the same to obtain US work permits for foreigners. (Harder);
- In the last few years, "the gap between minority and white students in math and reading tests at the elementary school level" has increased, decreased, or stayed about the same. (Increased);
- Number of uninsured Americans. Answer choices: 1 million, 25 million, 45 million, 65 million or 85 million.(45mill);
- "Over the next fifty years, do you think the number of Americans over age 65 will..." Answer

choices: tripled, doubled, increased by half, or stayed about the same (Doubled);

- The percentage of uninsured low income families. Answer choices: a quarter, fifty percent, two-thirds, or ninety percent (Two Thirds)

D.14 Bulgaria

(All of the questions were True or False)

- If a policeman wants to detain a citizen for questioning at any time of the day or night, the citizen does not have the right to refuse()
- The Court has the right to detain suspects as long as needed to prove if they are guilty or not()
- The prosecution can detain a suspect until the case is resolved()
- The Chief Prosecutor is responsible and accountable to Parliament()
- The police do not have the right to use violence against a detainee even if proven that he/she has committed a crime()
- The police can detain you for 72 hours()
- A confession is not enough to find the defendant guilty()

D.15 Texas Utility Poll (CPL)

- What energy source do you think produces the greatest amount of CPL's electricity; coal, wind, natural gas, fuel oil, nuclear, or solar? (Natural Gas)
- Overall, what group of CPL customers consume the most kilowatt hours of electricity; residential, commercial, or industrial? (Industrial)
- Which customer group pays the highest electric rates; residential, commercial, or industrial? (Commercial)
- Which of the following resources used to produce electricity produces the most air emissions; nuclear, coal, or natural gas? (Coal)
- Roughly, how much profit do Texas electric utilities make on fuel; 0%, 10%, 20%, or 30%? (0%)
- Which agency sets electric rates in Texas? – PUC, Texas Utility Commission, Government, CPL, WT, SWEPCO, Other. (PUC)
- How much of your electric bill goes for fuel; less than 20%, 21 to 30%, 31 to 40%, 41 to 50%, or more than 50%? (21 – 30%)